Costain Nachuma

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Professional Summary

PhD candidate in Computer Science (4.0 GPA) at Idaho State University's SECReT Lab, specializing in empirical analysis of AI agent behavior, LLM usage patterns, and software supply chain security. Eight peer-reviewed publications at MSR, ACM SAC, Springer ITNG, and ACM CompEd with 107+ citations. Experienced in building large-scale data pipelines, applying statistical modeling, and translating complex datasets into actionable findings. Sole instructor of record for a full undergraduate course. Seeking data science and ML engineering roles where rigorous analysis meets real-world impact.

Technical Skills

Languages	Python, SQL, Java, C#, JavaScript
Data & ML	Pandas, NumPy, Statistical Modeling (Linear/Logistic Regression), Feature Engineering, Data Preprocessing
LLM & Agents	LLM behavior analysis, prompt engineering, agent benchmarking, empirical evaluation of AI-assisted workflows
Databases	PostgreSQL, MySQL, Oracle, Neo4j (Graph Databases)
DevOps & Systems	Linux, Git, Docker, GitHub Actions
Research Methods	Empirical Analysis, Systematic Mapping Studies (Kitchenham), Reproducible Pipelines, HCI Methods
Cybersecurity	Software Supply Chain Risk Analysis, Threat Intelligence, SBOM Evaluation

Professional Experience

Graduate Research & Teaching Assistant · *Idaho State University*, Pocatello, ID — Aug 2023 – Present

- Built reproducible end-to-end data pipelines extracting and modeling pull request metadata and dependency graphs from GitHub and Maven at scale using Python (Pandas, NumPy, DuckDB).
- Applied logistic regression with repository-clustered standard errors to analyze AI agent collaboration patterns across 33,596 agent-authored pull requests; findings published at MSR 2026.
- Performed large-scale empirical analysis of LLM usage in software development workflows, characterizing ChatGPT-assisted coding behavior across real-world projects; published at MSR 2024.
- Conducted quantitative vulnerability analysis across the Maven dependency ecosystem, identifying risk propagation patterns across large-scale dependency graphs; published at MSR 2025.
- Designed and implemented an Eclipse IDE plugin for interactive software visualization, enabling developers to explore structural relationships and metrics directly within the development environment.
- Conducted a Systematic Mapping Study (Kitchenham's guidelines) synthesizing 100+ papers on software visualization tools.
- Mentored 20+ graduate and undergraduate students in advanced OOP, algorithms, and database systems.

Instructor of Record — CS 3309: Advanced Object-Oriented Programming
Idaho State University, Pocatello, ID — Fall 2025

- Sole instructor for a full undergraduate course covering advanced OOP principles, design patterns, and software engineering practices.
- Independently designed coursework, delivered all lectures, held office hours, and assessed student performance for a full semester.

Senior ICT Officer / Software Engineering Lead
National Assembly of Zambia, Lusaka, Zambia — Jun 2012 – Sep 2023

- Directed software development and IT operations supporting 300+ users across government departments; maintained high system reliability for mission-critical parliamentary systems.

- Reduced issue resolution time by 30% through systematic workflow analysis and infrastructure optimization.
- Led end-to-end development and modernization of parliamentary web platforms and digital workflow systems; managed sectional training budgets and technical training programs.

Systems Librarian Intern · *Prosentient Systems Pty*, Sydney, Australia May 2019 – Jul 2019

- Built a Java-based real-time communication tool improving collaboration across research institutions.
- Administered Linux servers and open-source repository platforms (DSpace, Openfire) using Git and SSH.

Education

Idaho State University Pocatello, ID

Ph.D. in Computer Science — Software Engineering & Cybersecurity | GPA: 4.0 Aug 2023 – May 2027

- **Dissertation:** Interactive IDE-integrated software visualization tools for code comprehension
- **Research areas:** Empirical SE, AI-assisted development, LLM behavior analysis, software supply chain security
- **Lab:** SECReT Lab (Software Engineering & Cybersecurity Research Team) — Advisor: Dr. Minhaz F. Zibran

University of New Orleans New Orleans, LA

M.Sc. in Computer Science 2021 – 2023

University of Zambia Lusaka, Zambia

B.Sc. in Computer Science 2007 – 2011

Selected Publications

When AI Teammates Meet Code Review: Collaboration Signals Shaping the Integration of Agent-Authored Pull Requests — *MSR 2026*
Nachuma, C. & Zibran, M. F.
23rd International Conference on Mining Software Repositories | doi.org/10.1145/3793302.3793561

Decoding Dependency Risks: A Quantitative Study of Vulnerabilities in the Maven Ecosystem — *MSR 2025*
Nachuma, C., Hossan, M. M., Turzo, A. K. & Zibran, M. F.
22nd International Conference on Mining Software Repositories

ChatGPT in Action: Analyzing Its Use in Software Development — *MSR 2024*
Champa, A. I., Rabbi, M. F., Nachuma, C. & Zibran, M. F.
21st ACM International Conference on Mining Software Repositories

SBOM Generation Tools Under Microscope: A Focus on the npm Ecosystem — *ACM SAC 2024*
Rabbi, M. F., Champa, A. I., Nachuma, C. & Zibran, M. F.
39th ACM/SIGAPP Symposium on Applied Computing

Future Ready, Globally Relevant: Aligning Africa's Tertiary CS Education with Global Standards — *ACM CompEd 2025*
Jonathan M., Oudshoorn M., Binamungu L. P., Bainomugisha E., Nachuma, C., et al.
ACM Global Computing Education Conference 2025

Additional publications at ITNG 2025 (Springer), i-ETC 2024/2026, and earlier venues. Full list: Google Scholar (107+ citations, 8 papers)

Awards & Recognition

- **Top Graduate Oral Presentation** — ISU Research & Creative Works Symposium, Engineering/Physical & Mathematical Sciences, 2026
- **Top Graduate Poster Award** — ISU Research & Creative Works Symposium, Engineering/Physical & Mathematical Sciences, 2026

- **Best Poster Award** — 4th Annual Intermountain Engineering, Technology & Computing Conference, 2024
- **Best Oral Presentation Award** — ISU Research & Creative Works Symposium, 2024

Leadership & Volunteering

- **Team Lead**, TechEquity AI Summit, Plug and Play Tech Center, Sunnyvale, CA (Nov 2025) — Led 50+ volunteers supporting 2,000+ attendees; coordinated exhibitor and executive stakeholder engagement.
- **CS Mentor**, Idaho State University (2023–Present) — Mentorship in programming, career development, and technical interview preparation for 20+ students.
- **President**, University of Zambia Computer Society (UNZACS), 2010 — Organized coding bootcamps and workshops.

Certifications

- IBM / Coursera: Python for Data Science, AI & Development (2026)